2.5 Intermolecular forces

Students will be assessed on their ability to:

a demonstrate an understanding of the nature of intermolecular forces resulting from interactions between permanent dipoles, instantaneous dipoles and induced dipoles (London forces) and from the formation of hydrogen bonds

b relate the physical properties of materials to the types of intermolecular force present, e.g.:

- i the trends in boiling and melting temperatures of alkanes with increasing chain length
- ii the effect of branching in the carbon chain on the boiling and melting temperatures of alkanes
- iii the relatively low volatility (higher boiling temperatures) of alcohols compared to alkanes with a similar number of electrons
- iv the trends in boiling temperatures of the hydrogen halides HF to HI

c carry out experiments to study the solubility of simple molecules in different solvents

d interpret given information about solvents and solubility to explain the choice of solvents in given contexts, discussing the factors that determine the solubility including:

- i the solubility of ionic compounds in water in terms of the hydration of the ions
- ii the water solubility of simple alcohols in terms of hydrogen bonding
- iii the insolubility of compounds that cannot form hydrogen bonds with water molecules, e.g. polar molecules such as halogenoalkanes
- iv the solubility in non-aqueous solvents of compounds which have similar intermolecular forces to those in the solvent.